

Single Sample Per Person Face Recognition Based on Deep Convolutional Neural Network

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Abstract—As for that there is only one training sample in single sample per person(SSPP) face recognition(FR), deep learning(DL) method is very difficult to be used. To overcome this problem, the paper proposes a series of methods to make it possible to be used in SSPP FR. Firstly, an expanding sample method is proposed to increase training sample. Secondly, a learned Deep Convolutional Neural Network(DCNN) model which is trained with amounts of face images and can represent face very well is brought in. Then, these expanding samples are used to fine-tune the DCNN model. Thirdly, the fine-tuned DCNN model is used to perform experiment. Experiments demonstrate that a better performance is obtained by using the fine-tuned DCNN model.

Keywords—single sample per person; face recognition; deep convolutional neural network; fine-tuning

I. INTRODUCTION

As the application of immigration management, fugitive tracing, and video surveillance, the technique of SSPP FR has become an important research topic in academic. However, since there is only one training sample, it is difficult to predict facial variations such as illumination, disguise, etc. with traditional method. In addition, as for lacking training sample, it is also impossible to use DL method. Both of these make SSPP FR more difficult to be done than other topics in computer vision(CV).

In recent years, lots of methods have been proposed to solve this problem. But most of these methods are traditional method. Generally speaking, these methods can be divided into four methods: expanding training sample method[1], generic learning method[2], patch based method[3] and extracting multi-scale descriptors method[4]. Expanding training sample method is to expand sample using some special measures such as photometric transform and geometric distortion[5], perturbation based method[6] and so on. Generic learning method is based on the fact that

facial variations can be shared across different persons and with the help of an extra face database to learn similar facial variations. Patch based method extracts features on several face patches respectively, then does classification. Extracting multi-scale descriptors method uses different filters to extract facial features, then does classification.

So far, there is few DL methods for SSPP FR. [7] proposed a DL method to solve SSPP FR problem via learning deep supervised auto-encoder. Firstly, a supervised auto-encoder enforces variation faces to be mapped with canonical face and enforces features of same face to be similar. Then, the supervised auto-encoder is stacked to obtain deep architecture. Finally, the deep supervised auto-encoder is used to extract features.

Inspired by [7] and driven by Artificial Intelligence technology, the paper proposes a method using DCNN to do SSPP FR. Firstly, an expanding sample method is proposed to expand training sample. Secondly, a learned DCNN model is brought in, then the model is fine-tuned by these expanding samples. Finally, the fine-tuned model is used to perform experiment. The framework of the proposed method is illustrated in Figure 1.

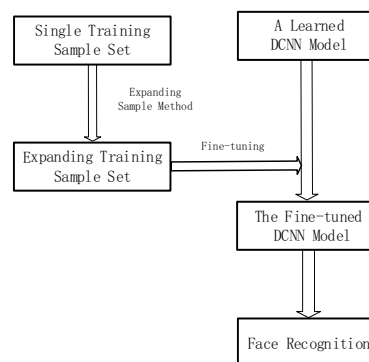


Figure 1. The framework of the proposed method.

II. THE EXPANDING SAMPLE METHOD

In order to overcome the lack of training sample in SSPP FR, the paper proposes an expanding sample method which is based on the fact that facial variations can be shared among different persons. Its basic principle diagram is illustrated in Figure 2.



Figure 2. The basic principle diagram of the expanding sample method.

The specific steps of the expanding sample method are summarized in Table I.

TABLE I. THE ALGORITHM OF THE EXPANDING SAMPLE METHOD

Input: an extra database and single sample Output: expanding samples
1. Calculate the bias of each person's every variation, as following: $\varepsilon_{ij} = X_{ij} - X_{i1};$
2. Calculate the average bias of each variation in the database, as following: $\bar{\varepsilon}_j = \frac{1}{m} \sum_{i=1}^m \varepsilon_{ij};$
3. Construct intra-class variation set by step 2, as following: $\bar{\varepsilon} = (\bar{\varepsilon}_1, \bar{\varepsilon}_2, \dots, \bar{\varepsilon}_n);$
4. Output expanding samples, as following: $E_k = X_k + \bar{\varepsilon}.$

a. In TABLE I, X_{ij} represents the j th variation of the i th person, X_{i1} represents the frontal neutral image of the i th person, ε_{ij} represents the bias of the i th person's j th variation corresponding to its neutral face, m is the number of person in the database, n is the number of each person's variation, $\bar{\varepsilon}_j$ represents the average bias of the j th variation, $\bar{\varepsilon}$ represents the intra-class variation set. X_k represents the neutral face image of the person k . E_k represents the expanding samples of the person k .

The framework of constructing intra-class variation set is shown in Figure 3.

III. DEEP CONVOLUTIONAL NEURAL NETWORK(DCNN)

A. Transfer Learning

Transfer Learning uses the knowledge learned from one specific scene to help other application scene. In other words, it uses associated data to learn a model or mapping, then uses the model or mapping to do new task.

Since there is one training sample in SSPP FR, DCNN which needs abundant data to be trained is difficult to be used. Therefore, having the aid of a lightened CNN[8] which can learn a compact embedding for face recognition, the research is done.

The lightened CNN introduces a new activated function named Max-Feature-Map which uses maxout in the fully connected layer to the convolution layer. Given an input

convolution layer $C \in R^{h \times w \times 2n}$, the Max-Feature-Map activation function can be written as follows:

$$f_{ij}^k = \max(C_{ij}^k, C_{ij}^{k+n}) \quad (1)$$

where the channel of the input convolution layer is $2n$, $1 \leq i \leq h$, $1 \leq j \leq w$.

The architecture of the lightened CNN is illustrated in Figure 4.

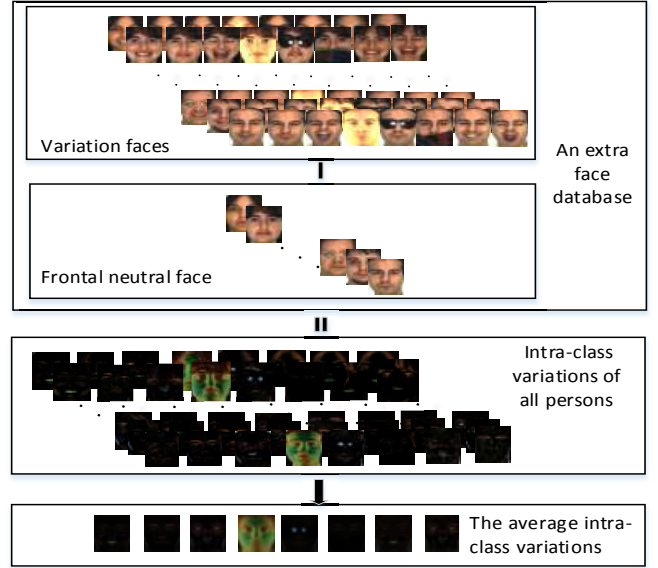


Figure 3. The framework of constructing intra-class variation set.

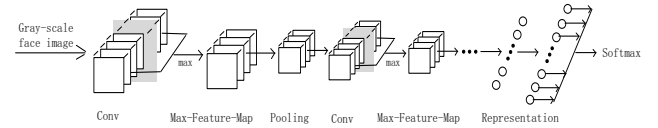


Figure 4. The architecture of the Lightened CNN.

B. Fine-tuning

The lightened CNN is trained by CASIA-WebFace database. The CASIA-WebFace database contains 10,575 persons and has a total of 493,456 face images. Before it is used to train the lightened CNN, it is firstly pre-processed. The pre-processing include that it is converted to gray-scale images and normalized to 144×144 . After pre-processed, it is used to train the lightened CNN. Later, a well-trained model is obtained. The well-trained model is used to do the fine-tuning task. The expanding samples are used to fine-tune the model. And the fine-tuned model is used to implement experiment.

IV. EXPERIMENT

AR face database includes 76 men and 50 women more than 4000 face images. A sub-database with 2600 face images is used here. It contains 100 persons including 50 men and 50 women, each person has 26 face images.

Single sample and expanding samples are used to fine-tune the DCNN model, respectively. Then these fine-tuned models are used to implement experiments, respectively. The FR rates of illumination, expression, disguise, and disguise with illumination variations in session 1 and session 2[2] are shown in TABLE II and TABLE III, respectively.

TABLE II. FACE RECOGNITION RATE(%) ON AR FACE DATABASE(SESSION 1)

Method	Illu	Exp	Dis	Disill
SRC[9]	80.8	85.4	55.6	25.3
CRC[10]	80.5	80.4	58.1	23.8
AGL[11]	93.3	77.9	70.0	53.8
DMMA[3]	92.1	81.4	46.9	30.9
PNN[12]	84.6	86.7	90.0	72.5
PCRC[13]	95.0	86.7	95.6	81.3
ESRC[14]	99.6	85.0	83.1	68.6
SVDL[15]	98.3	86.3	86.3	79.4
LGR[2]	100	97.9	98.8	96.3
TLC[4]	100	98.3	99.4	98.1
Single training sample(100)	99.3	94.8	95	87.3
The proposed method	100	95	96.5	88.3

b. In TABLE II and TABLE III, Illu represents illumination variation, Exp represents expression variation, Dis represents disguise variation, and Disill represents disguise with illumination variation.

TABLE III. FACE RECOGNITION RATE(%) ON AR FACE DATABASE(SESSION 2)

Method	Illu	Exp	Dis	Disill
SRC[9]	55.8	68.8	29.4	12.8
CRC[10]	55.8	69.6	35.0	13.5
AGL[11]	70.8	55.8	40.6	30.7
DMMA[3]	77.9	61.7	28.1	21.9
PNN[12]	77.5	73.8	71.9	52.8
PCRC[13]	88.8	71.7	81.8	63.1
ESRC[14]	87.9	70.4	59.4	45.0
SVDL[15]	87.1	74.2	61.3	54.1
LGR[2]	97.5	85.0	93.8	88.8
TLC[4]	99.2	87.1	96.3	91.9
Single training sample(100)	97.3	88.1	86	72.8
The proposed method	99.8	88	86	76.3

It can be seen from TABLE II that the recognition rate of the proposed method is slightly less than the first two, but it outperforms the third by 1.7%, 8.7%, 10.2%, 8.9% under different variation conditions, respectively. However, the recognition rate of the fine-tuned model that obtained by

using single sample to fine-tune the DCNN model outperforms the third by 1%, 8.5%, 8.7%, 7.9% under different variation conditions, respectively. It also can be seen from TABLE II that the recognition rates of the fine-tuned model that obtained by using single sample to fine-tune the DCNN model are lower than those of the fine-tuned model that obtained by using the proposed method by 0.7%, 0.2%, 1.5%, 1% under different variation conditions, respectively.

From TABLE III, it can be clearly find that the illumination recognition rate and expression recognition rate of the fine-tuned model that obtained by using the proposed method outperform the second by 0.6% and 0.9%, respectively. The recognition rates of illumination and disguise with illumination of the fine-tuned model that obtained by using the proposed method outperform those of the fine-tuned model that obtained by using single sample to fine-tune the model by 2.5% and 3.5%, respectively.

TABLE IV. THE FR RATE(%) ON AR FACE DATABASE(ALL)

Method	AR
DAE	37.6
MDAE	80.63
SN	81.15
AGL	58.86
CRC	54.16
SRC	54.99
OSS	78.75
LBP	80.83
ESRC	81.25
SSAE[7]	85.21
Single training sample(100)	92
Expanding sample(2600)	92

c. the data in TABLE IV except that obtained in the experiment are from paper[7].

The recognition rates on AR face database of the fine-tuned model that obtained by using single sample to fine-tune the model and the proposed method respectively are shown in TABLE IV. It is clearly that the proposed method has a outstanding performance among these methods. And the experimental result of the proposed method outperforms the second by 6.79%.

It can be easily found that in TABLE III, the expression recognition rate of the fine-tuned model that obtained by using single sample to fine-tune the DCNN model is higher than that of the fine-tuned model which is obtained by using the proposed method, the disguise recognition rate of the fine-tuned model that obtained by using single sample to fine-tune the DCNN model has the same accuracy as that of the fine-tuned model which is obtained by using the proposed method and the recognition rate of the fine-tuned model that obtained by using single training sample to fine-

tune the DCNN model is equal to that of the fine-tuned model which is obtained by using the proposed method in TABLE IV. To solve the doubt, these losses of the fine-tuned model that obtained by using single sample to fine-tune the DCNN model and the proposed method are tested, respectively. As shown in TABLE V.

As shown in TABLE V, it can be seen that in session 1, the losses of the fine-tuned model that obtained by using single sample to fine-tune the DCNN model is 48.555, 1.346, 1.374 and 1.446 times as much as those of the fine-tuned model that obtained by using the proposed method under illumination, expression, disguise, and illumination with disguise variation conditions, respectively. In session 2, the losses of the fine-tuned model that obtained by using single sample to fine-tune the DCNN model is 5.749, 1.138, 1.401 and 1.325 times as much as those of the fine-tuned model that obtained by using the proposed method under different variation conditions, respectively. All in all, the results show that these losses of the fine-tuned model that obtained by using the proposed method are less than those of the fine-tuned model that obtained by using single training sample to fine-tune the DCNN model. It demonstrates the proposed method makes a difference for SSPP FR.

TABLE V. THE LOSSES OF USING SINGLE SAMPLE AND EXPANDING SAMPLES TO FINE-TUNE THE MODEL

Testing Set	Fine-tuning Sample	Sample	
		Single Sample	Expanding Sample
Session 1	Illumination	0.0464923	0.000957526
	Expression	0.36872	0.274005
	Disguise	0.252945	0.184081
	Illumination+Disguise	0.815179	0.563917
Session 2	Illumination	0.114036	0.0198348
	Expression	0.688221	0.604769
	Disguise	0.707796	0.5052
	Illumination+Disguise	1.46812	1.10823

V. CONCLUSION

In the paper, a new approach is proposed to solve the face recognition(FR) problem that with single sample per person(SSPP). Firstly, a novel expanding sample method for SSPP FR is proposed. Secondly, a well-trained DCNN is brought in, then the expanding samples are used to fine-tune the DCNN model. Finally, AR face database is used to test the accuracy of SSPP FR on the fine-tuned DCNN model. Experimental results show that the proposed method achieve some higher recognition rates of illumination and expression than the second in session 2 and also achieve the highest accuracy on all images of AR face database. Besides, the losses of session 1 and session 2 of the fine-tuned model that obtained by using single sample to fine-tune the DCNN model and the proposed method are tested, respectively. The

results demonstrate that the proposed method has a better performance on SSPP FR compared with that using single sample to fine-tune the DCNN model. Besides, the paper becomes a pioneer that use DCNN in SSPP FR and demonstrates that DCNN can be used in the task who has single or few training samples. A research on SSPP FR with DCNN will be continued in the future.

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